1. Project Overview

This data analytics project explores musical participation in Japan while also examining the distribution of schools across Japanese prefectures. The findings related to the school system will serve as a foundation for a future, related research project.
The primary data source for this analysis is Japan's 2021 Survey on Time Use and Leisure Activities, published through the official e-Stat statistics portal.
Before conducting the analysis, seven Japanese cross-tabulation tables were translated into English with AI assistance. This was followed by an initial exploration of the datasets to identify relevant variables, understand the available categories, and determine which information would be most useful for the research questions.
A key part of the preparation involved creating clean, structured datasets from the original statistical tables. Relevant categories were selected, and CSV formatting and missing-value representations were standardized to ensure consistent data types and reliable import into Google BigQuery.
These preparation steps established the foundation for the subsequent analysis of musical participation across Japan.

2. Business Question

Music in Japan: Exploring Musical Participation and School Band Culture

- *How does musical participation vary across Japan, and what demographic, cultural, and educational patterns can be identified?*

3. Dataset and Selection

Japan's 2021 Survey on Time Use and Leisure Activities, published through the official e-Stat statistics portal and School Basic Survey 2025.

4. Tools Used

- Excel
- BigQuery SQL
- Power BI

5. Data Preparation / Cleaning

Seven Japanese cross-tabulation tables were selected from the official e-Stat database and translated into English with AI assistance.

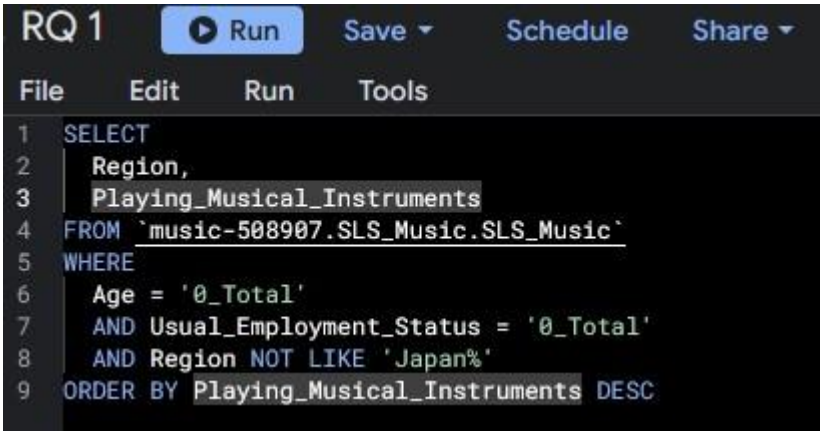
Afterwards, the datasets were examined to understand their structure and identify relevant categories and variables for the research questions.

During data cleaning, CSV formatting, missing values, and data types were standardized to ensure a consistent structure and reliable import into Google BigQuery.

Finally, the relevant data was selected and prepared for further analysis using SQL and Power BI.

RQ1 – Regional perspective

Which prefecture in Japan has the highest participation rate in playing musical instruments?




```sql
SELECT
  Region,
  Playing_Musical_Instruments
FROM `music-508907.SLS_Music.SLS_Music`
WHERE
  Age = '0_Total'
  AND Usual_Employment_Status = '0_Total'
  AND Region NOT LIKE 'Japan%'
ORDER BY Playing_Musical_Instruments DESC
```

Tokyo has the highest participation rate in playing musical instruments among Japanese prefectures, at 13.8%. Kanagawa ranks second at 11.2%, followed by Kyoto at 10.6%.

RQ2 – Musical perspective

How does participation in classical and popular music concerts relate to playing musical instruments across Japan?

```sql
SELECT
  CORR(Popular_Music_Concerts, Playing_Musical_Instruments) AS Popular_Coeff,
  CORR(Classical_Concerts, Playing_Musical_Instruments) AS Classical_Coeff
FROM `music-508907.SLS_Music.SLS_Music`
WHERE
  Age = '0_Total'
  AND Usual_Employment_Status = '0_Total'
  AND Region NOT LIKE 'Japan%'
```

(Query_COEFF)

```sql
SELECT
  Classical_Concerts,
  Popular_Music_Concerts,
  Playing_Musical_Instruments,
  Region
FROM `music-508907.SLS_Music.SLS_Music`
WHERE
  Age = '0_Total'
  AND Usual_Employment_Status = '0_Total'
  AND Region NOT LIKE 'Japan%'
ORDER BY Playing_Musical_Instruments DESC
```

(Query_Playing Musical Instruments)

Both classical and popular music concert participation show a positive relationship with playing musical instruments across Japanese prefectures. The relationship is somewhat stronger for popular music concerts (r = 0.71) than for classical music concerts (r = 0.64).

RQ 3 – People perspective

How does musical instrument participation vary by age and employment status in Japan?

```sql
SELECT distinct
  Age
FROM `music-508907.SLS_Music.SLS_Music`
```

(Query_Age Distinct)

```sql
SELECT distinct
  Usual_Employment_Status
FROM `music-508907.SLS_Music.SLS_Music`
```

(Query_Employment Distinct)

```sql
SELECT distinct
  Age,
  Usual_Employment_Status,
  Playing_Musical_Instruments
FROM `music-508907.SLS_Music.SLS_Music`
WHERE
  Age IN (
  '1_15-24 years',
  '2_25-34 years',
  '3_35-44 years',
  '4_45-54 years',
  '5_55-64 years',
  '6_65-74 years',
  '7_75 years and over'
)
AND
  Usual_Employment_Status IN (
    '11_Mainly working',
    '12_Working while mainly doing housework etc.',
    '13_Working while attending school',
    '21_Housework',
    '22_Attending school',
    '23_Other'
  )
AND
  Region IN ('Japan (Total)')
AND
  Playing_Musical_Instruments IS NOT NULL
ORDER BY Playing_Musical_Instruments DESC
```

(Query_Participation Distinct)

Higher musical instrument participation rates are concentrated among younger age groups, particularly those attending school or working while attending school. The highest rate was observed among 15–24-year-olds working while attending school (23.5%),

followed by those attending school (22.3%). Housework-related groups also appear frequently among the highest participation rates.

RQ4 - School Landscape

How does the ratio of junior high schools to high schools vary across Japan?

```
File    Edit    Run    Tools
1  SELECT
2    Prefecture,
3    Junior_High_Schools_Total,
4    High_Schools_Total,
5    ROUND(Junior_High_Schools_Total / High_Schools_Total, 2) AS School_Ratio
6    FROM `music-508907.School.School`
7    ORDER BY School_Ratio DESC
```

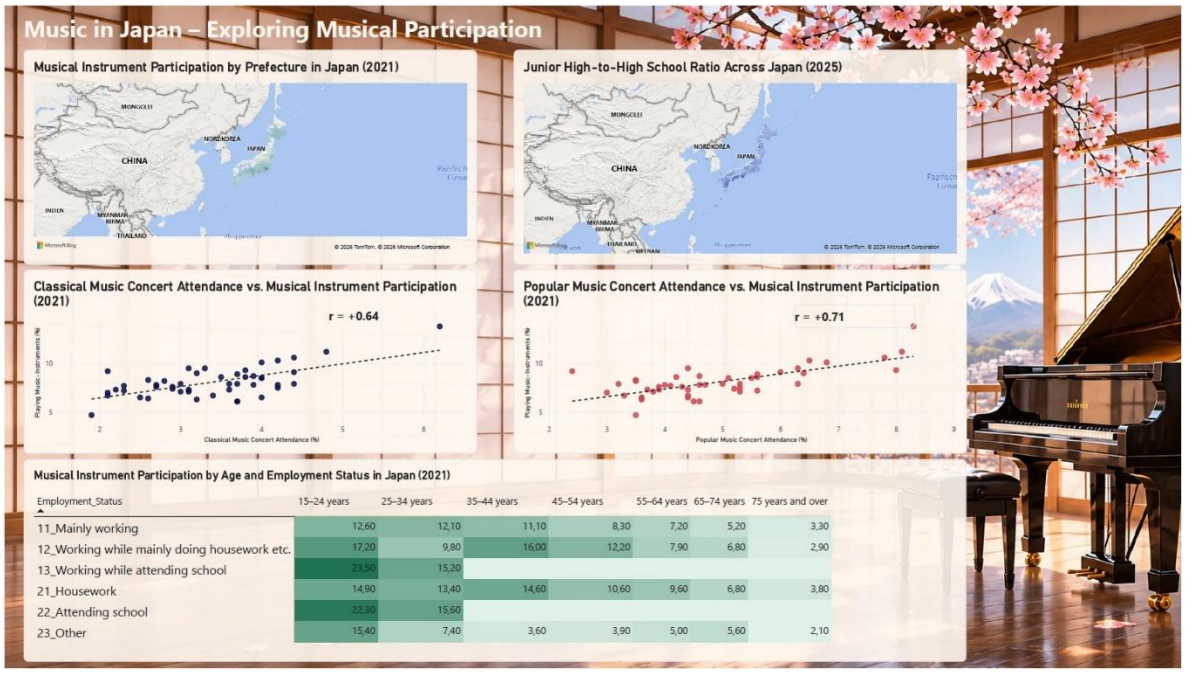
(Query Schools Ratio)

```
1  SELECT
2    Prefecture,
3    Junior_and_High_Schools_Total
4  FROM `music-508907.School.School`
5  ORDER BY Junior_and_High_Schools_Total DESC
```

(Query_Schools Total)

Tokyo has the highest combined number of junior high and high schools in Japan (1,226). However, the balance between school types reveals a different geographical pattern. Kochi has the highest junior-high-to-high-school ratio (2.77), followed by Wakayama (2.67) and Miyazaki (2.57). These findings highlight regional differences in Japan's school structure.

7. Dashboard



8. Key Findings / Conclusion

The analysis shows that younger age groups, particularly students, have the highest participation rates in playing musical instruments, followed by certain groups engaged in housework-related activities.

Furthermore, the results indicate a positive correlation between musical instrument participation and attendance at both classical and popular music concerts. The correlation is slightly stronger for popular music ($r = +0.71$) than for classical music ($r = +0.64$).

One possible explanation is that younger people, particularly students, may be more interested in popular music. However, this remains a hypothesis and cannot be confirmed using the available data.

The findings also suggest that schools may play an important role in introducing young people to musical instruments and encouraging musical participation. Nevertheless, the present analysis does not establish a direct relationship between school education and musical participation.

To investigate this possibility further, a separate analysis focusing on musical activities in Japanese junior high schools and high schools is planned.

This additional research may provide valuable insights into regional differences in school-based musical activities and their relationship to Japan's geographical and educational structure.